Predicting Facility-Level Greenhouse Gas Emissions Intensity in Alberta's Oil Sands Using Machine Learning

Paul Moynihan, Ifeanyi Njoku, Anmol Sharma

DATA 606 | Final Project

12th February, 2026

Executive Summary

Project Objective

Develop machine learning models to predict emissions intensity (tonnes CO_2e/m^3) at Alberta oil sands facilities, using only information that would legitimately be available for prediction — enabling better environmental compliance and emissions-reduction strategies.

Dataset Overview

- Facility-year records: 344 (2011–2023, after removing non-operational records)
- Target variable: Emission Intensity (tonnes CO_2e/m^3)
- Range: 0.184 – 7.433 | Average: 0.622
- Features: production volume, extraction technology, subsector, product, facility production history, time

Key Results

- Best model: Random Forest — Test $R^2 = 0.496$, RMSE = 0.741
- Top predictors: Log-transformed production (0.355) and raw production (0.259)
- 6 models compared: Linear Regression, Elastic Net, Random Forest, Gradient Boosting, SVR, Neural Network
- Production scale explains ~61% of feature importance — the dominant driver

Note on this revision: An earlier draft reported $R^2 = 0.681$ using features (Cogen_Adjusted_Emission, Log_Emission) mathematically entangled with the target, and a separate draft reported $R^2 = 0.058$ using raw Facility identity as a predictor, which caused tree models to memorize rather than generalize. This deck uses a corrected, leakage-free feature set throughout.

Problem Statement & Business Context

The Challenge

- Industrial facilities must monitor and reduce emissions intensity
- Current emissions tracking is reactive, not predictive
- High variability across facilities and extraction technologies
- Need early indicators of high-intensity operations

Business Impact

- Regulatory compliance (carbon pricing, emissions caps)
- Cost optimization (reduce emissions tax burden)
- Operational efficiency improvements
- Environmental sustainability goals

Our Approach

- Predictive modeling using only information available before operations complete
- Careful feature engineering that avoids leaking the target into the inputs
- Comparison of six ML algorithms on interpretability, accuracy, and generalization
- Feature importance analysis to guide emissions-reduction strategy

Specific Objectives

Predict Emissions Intensity

Develop accurate facility-level predictions of GHG emissions intensity using only legitimate operational and technological data.

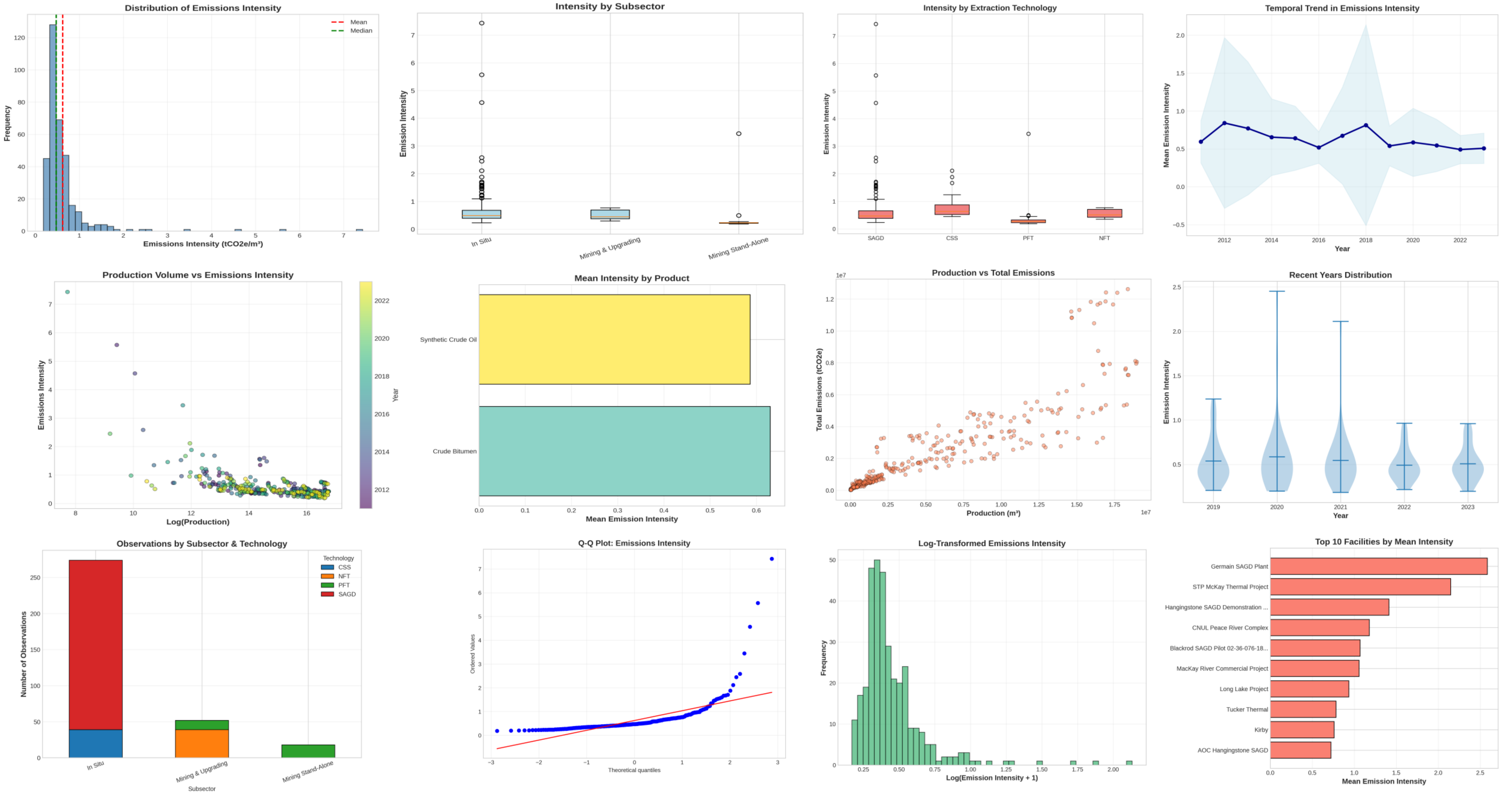
Compare ML Models

Evaluate six machine learning approaches across interpretability, accuracy, and generalization performance.

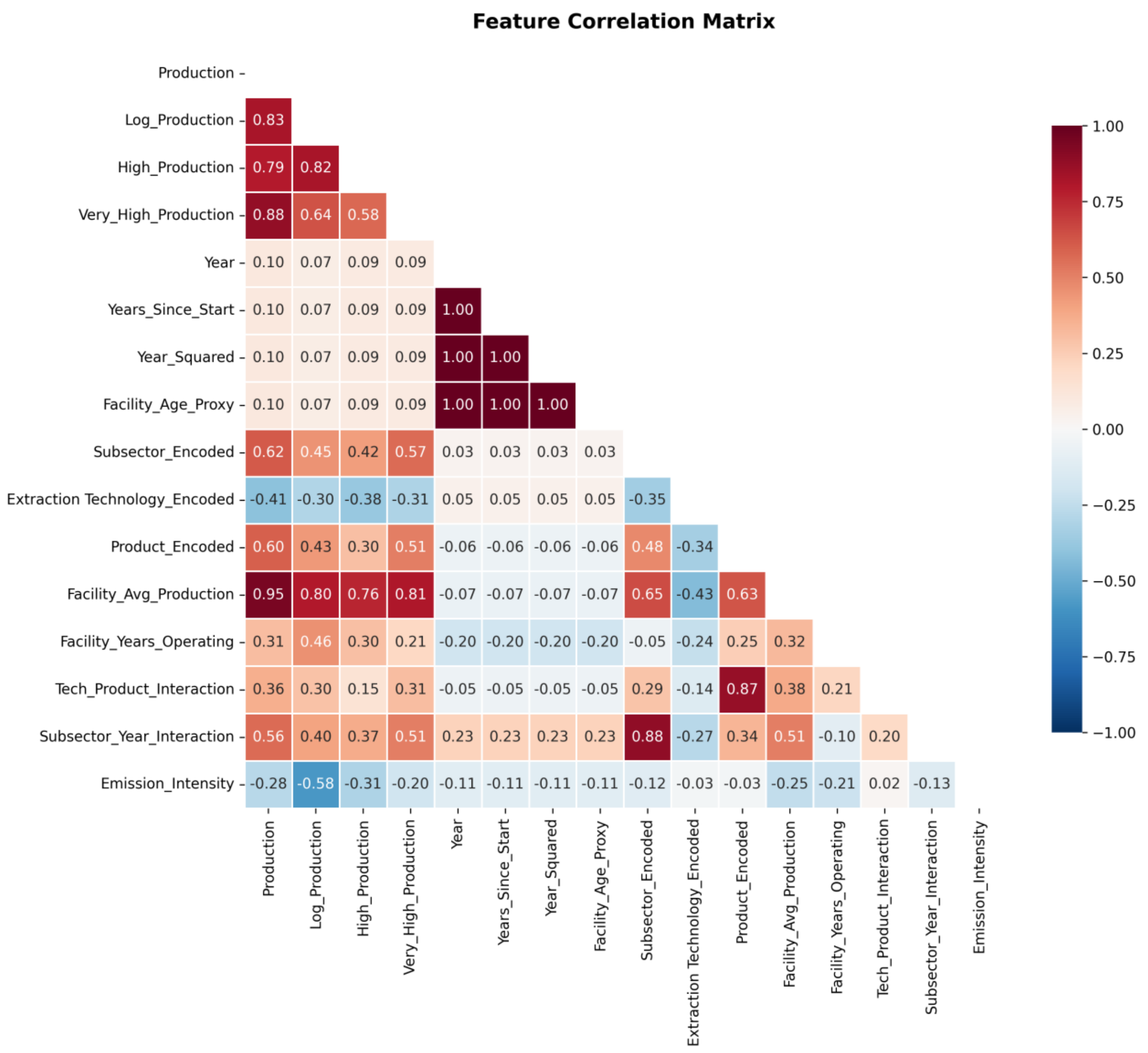
Identify Key Drivers

Determine which operational factors most significantly and defensibly impact emissions intensity.

Exploratory Data Analysis: Key Insights



Exploratory Data Analysis: Feature Correlations



What the correlations show

Production and Log_Production are negatively correlated with intensity — larger facilities tend to be more efficient (economies of scale)

Facility_Avg_Production behaves similarly — a facility's production history is informative even without using its identity

Note: Cogen_Adjusted_Emission and Log_Emission are excluded from the model feature set, since they are mathematically entangled with the target

Data & Methodology

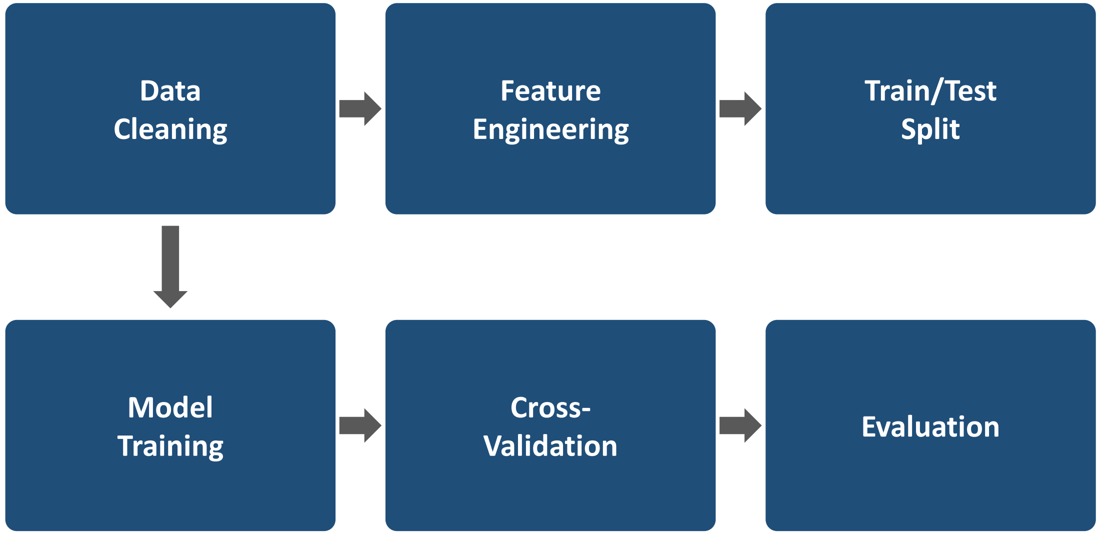
Variables

- Production volume (m^3)
- Total (cogen-adjusted) emissions — used only to compute the target
- Subsector, extraction technology, product type
- Facility and year

Leakage-Free Feature Engineering

- Log transformations of production
- Production-scale indicators (median / 75th percentile)
- Encoded categorical variables (subsector, technology, product)
- Facility-level historical aggregates (avg. production, years operating) — not facility identity
- Interaction terms (technology × product, subsector × time)

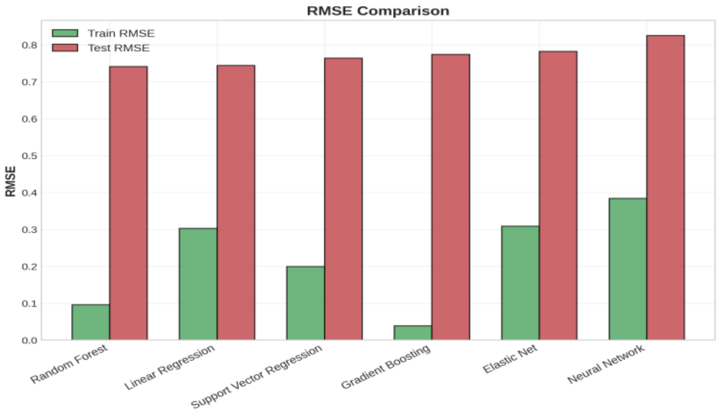
Machine Learning Pipeline

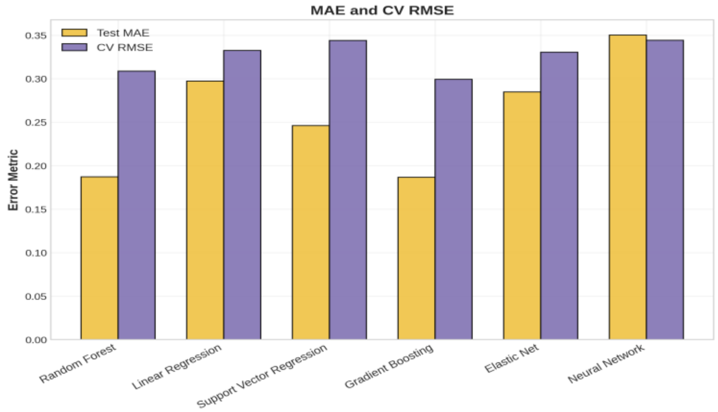


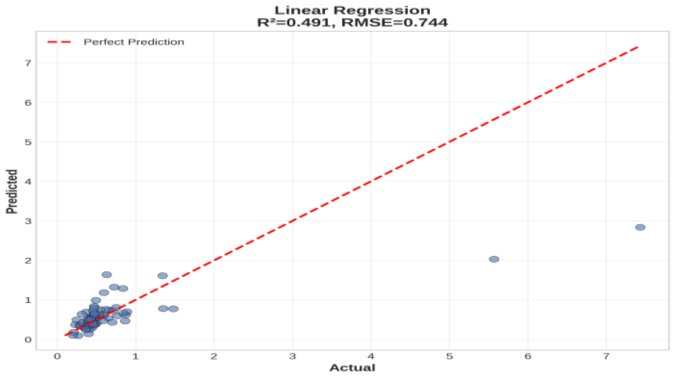
Machine Learning Models Compared

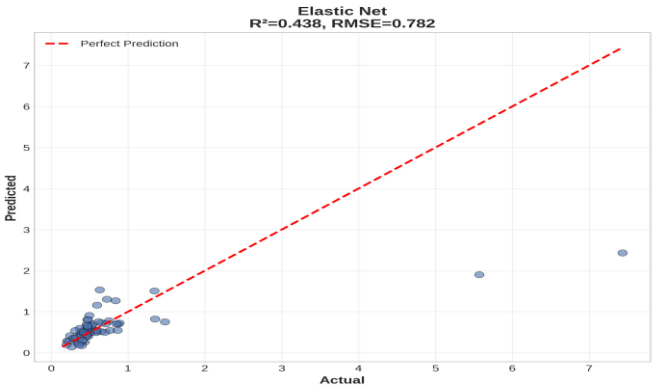
1. Linear Regression	Baseline linear model.	✓ Fast, interpretable, assumes linear relationships
2. Elastic Net	Combined L1/L2 regularization for feature selection.	✓ Handles multicollinearity, automatic feature selection
3. Random Forest	Ensemble of decision trees with bagging.	✓ Captures non-linearity, robust to outliers
4. Gradient Boosting	Sequential tree building with boosting.	✓ High accuracy, handles complex interactions
5. Support Vector Regression	Kernel-based regression with epsilon-insensitive loss.	✓ Effective in high dimensions, robust to outliers
6. Neural Network (MLP)	Multi-layer perceptron with early stopping.	✓ Captures complex non-linear relationships

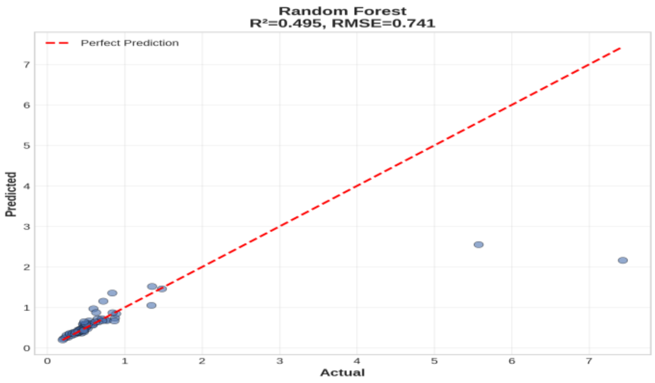
Model Comparison

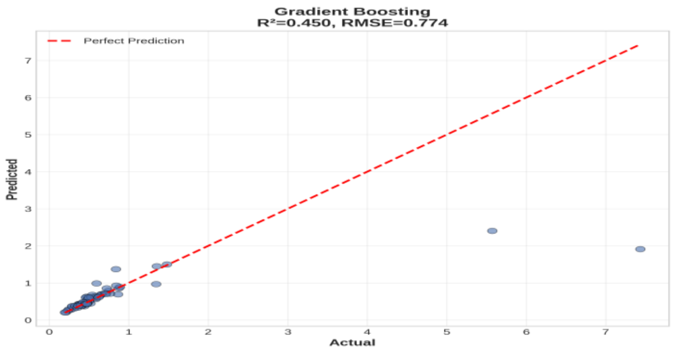


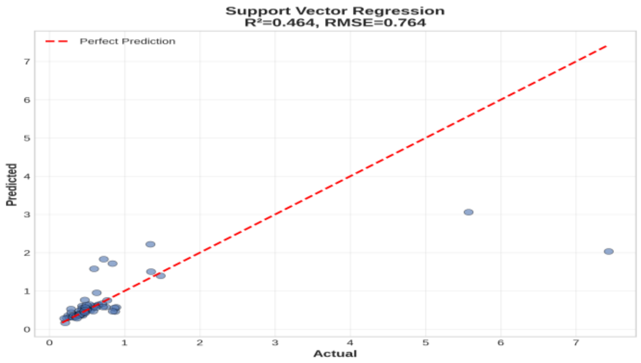


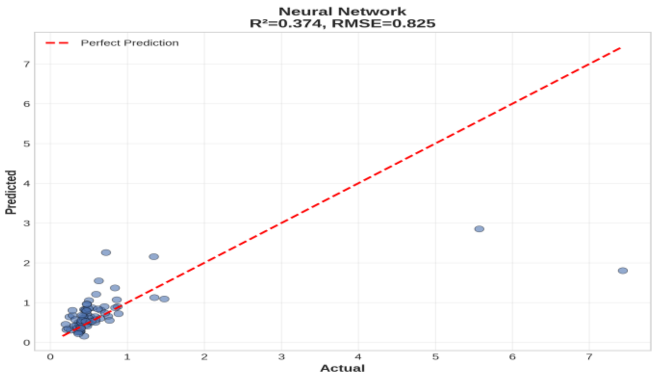










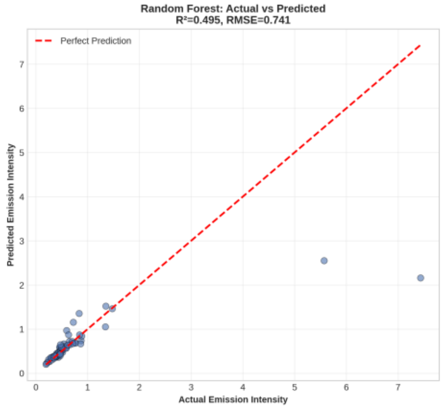


Model Performance Results

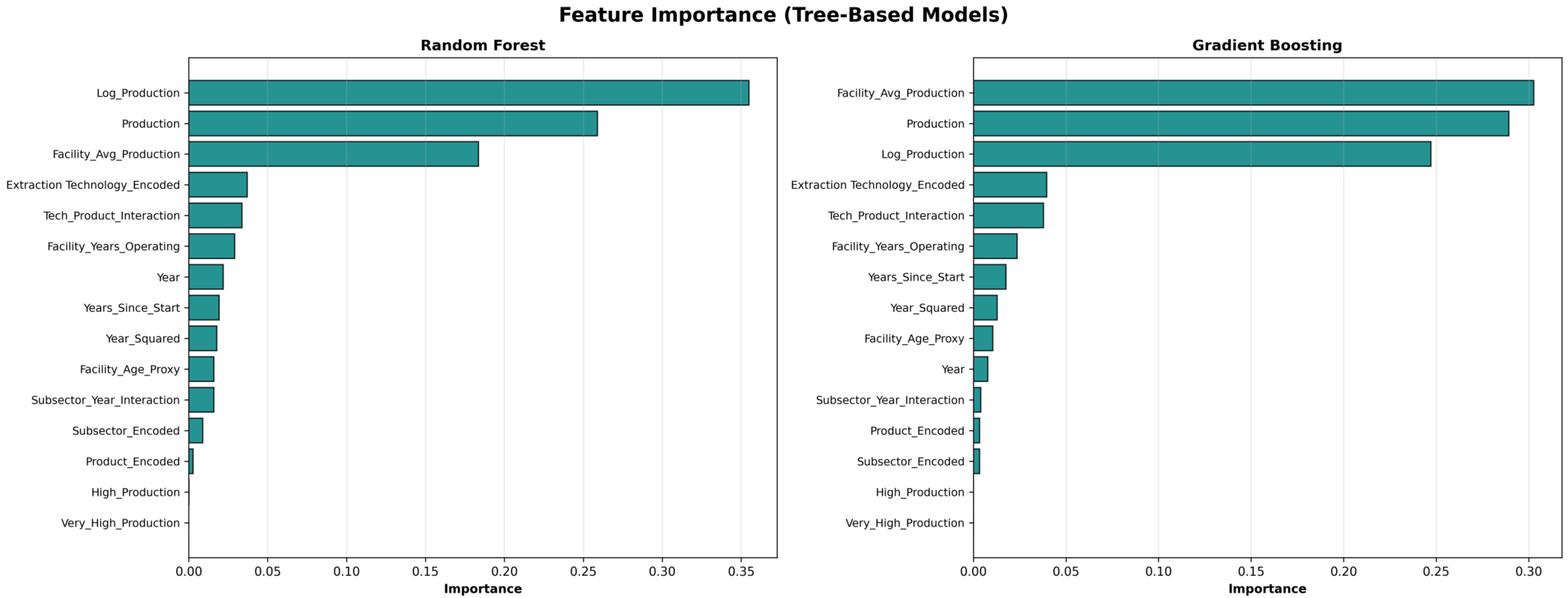
Model	Train R^2	Test R^2	Test RMSE	Test MAE	CV RMSE
Random Forest	**0.9558**	**0.4955**	**0.7410**	**0.1871**	**0.3087**
Linear Regression	0.5571	0.4913	0.7440	0.2973	0.3324
Support Vector Regression	0.8082	0.4641	0.7637	0.2461	0.3439
Gradient Boosting	0.9929	0.4498	0.7737	0.1866	0.2992
Elastic Net	0.5388	0.4376	0.7823	0.2847	0.3303
Neural Network	0.2868	0.3745	0.8250	0.3501	0.3442

★ **Best Model:** **Random Forest | Test R^2 = 0.4955 | RMSE = 0.7410 | MAE = 0.1871**

Random Forest achieves the best held-out performance, narrowly ahead of Linear Regression — both explain roughly half the variance using only production, technology, and temporal information.



Model Comparison – Feature Importance



- Log_Production (0.355) and Production (0.259) together explain ~61% of Random Forest importance — production scale is the dominant driver
- Facility_Avg_Production (0.184) — a facility's production history matters, without needing to know which facility it is
- Extraction Technology and interaction terms contribute smaller, secondary effects (~7% combined)

Key Findings & Insights

Statistical Insights

- Best model explains 49.6% of variance in emissions intensity (Test R^2 = 0.496)
- Average prediction error: 0.187 tonnes CO_2e/m^3 (Test MAE)
- Emission intensity range: 0.184 – 7.433 tonnes CO_2e/m^3 (40× variation)
- 344 facility-year records analyzed, 2011–2023

Key Drivers of Emissions Intensity

- Production scale (~61%): larger operations are systematically more efficient — economies of scale
- Facility production history (~18%): experienced, historically high-output facilities perform better
- Extraction technology (~7%): SAGD, CSS, NFT, PFT differ meaningfully but secondarily

Operational Implications

Focus emissions-reduction efforts on: (1) production-process optimization at scale, (2) sharing best practices from historically high-performing facilities, and (3) targeted evaluation of extraction technology upgrades where scale gains have been exhausted.

Model Diagnostics & Validation

Overfitting Analysis (Train R^2 – Test R^2)

⚠ Gradient Boosting: gap = 0.543 (severe)

⚠ Random Forest: gap = 0.460

⚠ Support Vector Regression: gap = 0.344

⚠ Elastic Net: gap = 0.101

⚠ Linear Regression: gap = 0.066 (smallest)

Why Random Forest Won

- Best test-set performance (Test R^2 = 0.4955)
- Lowest test prediction errors: RMSE (0.7410) among all six models
- Narrowly ahead of Linear Regression despite a larger train-test gap
- Captures non-linear interactions among production, technology, and time

Model Limitations

- All models show some overfitting (train performance exceeds test performance)
- Roughly half of the variance in emission intensity remains unexplained
- Missing variables: equipment age, maintenance schedule, energy source mix, carbon capture rates
- Facility-year records are treated as independent; no panel/time-series structure is exploited

Recommendations & Next Steps

Immediate Actions

- Use Random Forest predictions for facility benchmarking, not automated deployment
- Flag facilities whose predicted intensity is far below their actual intensity for review
- Report R^2 and RMSE together with the feature set used, to keep results reproducible

Operational Improvements

- Prioritize production-process optimization at scale
- Benchmark high- vs. low-intensity facilities within the same technology group
- Share best practices from historically efficient, high-production facilities

Future Model Enhancements

- Collect equipment specs, maintenance records, energy source, and CCS deployment data
- Apply panel-data methods (fixed/random effects) to exploit repeated facility-year structure
- Re-test facility identity only via random-effects terms, not as a raw leakage-prone predictor

Methodology Note: Correcting Target Leakage

The issue

Emission_Intensity = Cogen_Adjusted_Emission ÷ Production. An earlier draft included Cogen_Adjusted_Emission (and its log transform) directly as model features. Because these are mathematically entangled with the target's numerator, the resulting $R^2 = 0.681$ did not reflect genuine predictive skill. A later draft removed those features but used raw Facility identity as a label-encoded predictor; with 32 facilities and 344 rows, tree models memorized per-facility averages rather than learning generalizable drivers, producing near-zero or negative test R^2 (0.058 best case).

The fix

- ✓ Removed all emissions-derived features (Cogen_Adjusted_Emission, Log_Emission, and any emission/production ratio)
- ✓ Replaced raw Facility identity with facility-level historical aggregates (avg. production, years operating) — informative without being an identity key
- ✓ Retained production volume, technology, subsector, product, and temporal features as legitimate predictors

Result: Test $R^2 = 0.4955$ (Random Forest) — lower than the inflated 0.681, far better than the underpowered 0.058, and defensible under scrutiny.

References

- Charpentier, A. D., Bergerson, J. A., & MacLean, H. L. (2009). Understanding the Canadian oil sands industry's greenhouse gas emissions. Environmental Research Letters, 4(1), 014005.
- Environment and Climate Change Canada. (2023). National inventory report: Greenhouse gas sources and sinks in Canada. Government of Canada.
- Hastie, T., Tibshirani, R., & Friedman, J. (2017). The elements of statistical learning: Data mining, inference, and prediction (2nd ed.). Springer.
- National Pollutant Release Inventory. (2024). Facility-reported greenhouse gas emissions data. Government of Canada.
- Alberta Energy Regulator. (2024). Oil sands operational and production data. Government of Alberta.
- Alberta Energy Regulator. (n.d.). About the AER. https://www.aer.ca/about-aer

THANK YOU

Paul Moynihan · Ifeanyi Njoku · Anmol Sharma